

With projection based AR, your imagination is the only limit. There is a lot of research going on in this field and with time, more and more applications would pour in. If you are really excited about how you can create something like that of your own, we have tips for you at the end of this book. For now, let us see the location based AR.

Location based AR

It would be an injustice not to mention this category when talking about AR. Location based augmented reality is one of most widely implemented applications of AR. The strongest force behind this is the easy availability of smartphones and the features that they provide in terms of location detection. Location based AR is mostly used to help travellers in their journey.

Location based AR in most cases is used for AR location browsers which help users discover interesting places within their current location. This method works by detecting the user's location and orientation by reading data from the mobile's GPS, digital compass and accelerometer and predicting where the user is looking; then adding related information on screen about the objects that can be seen from the camera. Wikitude is one such application. If you've never used the app fear not, this video would help you realize how it looks in action: <http://dgit.in/WkXxwS> There are other interesting uses of location based AR which we will talk about in the coming chapters.

Outlining AR

Though the human eye is known to be the best camera in the world, there are limitations. We cannot look at things for too long. We cannot see well in low light conditions and sure as anything, your eye cannot see in infrared. For such cases, special cameras were built. Augmented reality apps which perform outlining use such cameras. Once again, object recognition sits behind all that outlining AR can do. Let us begin with a life-saving implementation example.

- 1 When driving a car on a road in foggy weather, the boundaries of the road may not be very visible to the human eye, leading to mishaps. Advanced cameras tuned specially to see the surroundings in low light conditions can be used to outline the road boundaries within which the car should stay. Such a system would prove very useful in avoiding accidents. With extra sensors capable of detecting objects around (e.g. by using ultrasound) the overall risk of hitting some living object can